Related Work: Historical Foundations, Active Inference Evidence Status, and Information Sources

Active Inference and FEP lineage: historical context and curriculum scope

Formal lineage and model commitments I

The FEP is commonly situated in a perception-as-inference lineage that includes Helmholtz, Bayesian-brain accounts of uncertainty, and tutorial treatments that translate the mathematics into worked learning sequences (Helmholtz 1867; Knill and Pouget 2004; Bogacz 2017; Sprevak and Smith 2023). Its modern variational formulation is developed in Friston's early and review papers (Friston et al. 2006; Friston 2010), and recent overview work usefully maps the breadth of related research without removing the need for local modeling commitments (Friston et al. 2023; Zhang and Xu 2024). Mathematical reviews and textbooks situate the principle as a variational bound on surprise and a process theory for perception, action, and learning (Buckley et al. 2017; Parr et al. 2022; Smith et al. 2022; Pezzulo et al. 2024). Associative-learning work makes

attention, prediction-error, blocking, and surprise phenomena explicit inside an Active Inference account, which is useful pedagogical context but not direct evidence for this curriculum's outcomes (Anokhin et al. 2024). The discrete POMDP formulation with A, B, C, and D matrices is the direct basis for the kernels here (K. Friston et al. 2017; Da Costa et al. 2020), while epistemic value and curiosity are treated by the expected-free-energy literature (Friston et al. 2015; K. J. Friston, Lin, et al. 2017; Millidge, Tschantz, et al. 2021; Sajid et al. 2021; Champion et al. 2026). Recent variational-inference and message-passing treatments make those formulation choices explicit by separating epistemic-prior, entropy-correction, and planning-correction assumptions rather than treating EFE as one unqualified objective (Nuijten, Lukashchuk, et al. 2026, 2025; Nuijten, Laar, et al. 2026).

Formal lineage and model commitments I

The planning-frontier argument therefore leans on those EFE planning and message-passing papers rather than on informal Lazy Dynamics talk material, because the manuscript needs source-backed claims about objectives, factorization, and assumptions rather than a weak conference-talk framing.

Ancient, medieval, and early-modern foundations of inference, perception, and sequenced learning

Ancient, medieval, and early-modern foundations of inference, perception, and sequenced learning I

This is a selective history of problems and design concerns, not a single-lineage account of Active Inference. The sources below are used to motivate an inspectable prerequisite order and to distinguish historical analogies from modern formal commitments. They do not show that modern FEP, Bayesian inference, variational inference, adaptive tutoring, or Active Inference existed before their later mathematical and computational formulations. The historical source set spans demonstration, optics, method, probability, association, education, and structured cognition (Aristotle 1901; Ibn al-Haytham 1989; Quintilian 95 AD; Bacon 1620; Descartes 1628; Huygens 1657; Bernoulli 1713; Moivre 1718; Bayes and Price 1763; Laplace 1774; Newton 1704; Berkeley 1709; Locke 1690, 1693; Hume 1739, 1748; Hartley 1749; Condillac 1754; Comenius 1657; Rousseau 1762; Kant 1781).

Demonstration, vision, and ordered instruction before modern probability I

Ancient demonstration and explanation provide one early vocabulary for separating reasons, evidence, and teachable order (Aristotle 1901). Quintilian's *Institutio Oratoria* presents a staged rhetorical education involving elementary studies, practice, revision, and progression through increasingly demanding forms of performance (Quintilian 95 AD). Its relevance here is historical: it shows that ordered instruction and repeated practice were explicit design concerns, not that ancient rhetoric implemented a learner model or adaptive tutor.

Demonstration, vision, and ordered instruction before modern probability I

The medieval Arabic optical tradition adds a distinct perception strand. Ibn al-Haytham's *Book of Optics*, represented here through a critical edition and translation, treats vision through geometrical analysis, observation, and experimentally constrained accounts of seeing (Ibn al-Haytham 1989). That tradition belongs in the history of perception and evidence, but it is not a predictive-coding theory, a Bayesian brain model, or evidence for the biological adequacy of the kernels in this release.

Early-modern method, probability, and experimental perception I

Early-modern method sources connect ordered reasoning to observation without collapsing their differences. Bacon's induction and experimental program, and Descartes's rules for moving through clear intuitions, deductions, and probable conjectures, make method and sequence explicit (Bacon 1620; Descartes 1628). Huygens's work on expectation in games of chance provides an early published probability treatment, while Bernoulli, de Moivre, Bayes, and Laplace extend the mathematical history of repeated observation and inverse-probability reasoning (Huygens 1657; Bernoulli 1713; Moivre 1718; Bayes and Price 1763; Laplace 1774). These sources support a history of calculation and inference under uncertainty, not a claim that their concepts were already variational or computational in the modern sense.

Early-modern method, probability, and experimental perception I

Newton's *Opticks* adds a controlled-experiment and sensory-observation strand: optical phenomena are varied, separated, and analyzed through explicit apparatus and mathematical argument (Newton 1704). Berkeley's account of vision similarly treats visual perception as involving learned relations among sensory cues, but neither source should be translated directly into modern generative models or neural predictive coding (Berkeley 1709).

Eighteenth-century association, education, and structured cognition I

Eighteenth-century sources then widen the educational and psychological context. Locke, Hume, Hartley, and Condillac discuss experience, association, habit, sensation, and reflection in different philosophical registers (Locke 1690; Hume 1739, 1748; Hartley 1749; Condillac 1754). Comenius, Locke's education writings, and Rousseau's *Émile* make staged progression, practice, motivation, and learner-sensitive ordering visible as historical pedagogical concerns (Comenius 1657; Locke 1693; Rousseau 1762). Kant provides a different account of structured cognition and recognition (Kant 1781). Together these sources justify treating sequence, representation, and evidence status as design questions; they do not establish modern cognitive science, personalized learning, or empirical efficacy.

Evidence-status critiques and empirical boundaries

Evidence-status critiques and empirical boundaries I

The same literature also warns against collapsing every layer of the project into a single grand claim. A variational identity, a process theory, a simulation recipe, and an empirical neuroscience hypothesis have different evidential burdens. Technical critiques of the FEP's particular-physics assumptions and subsequent Bayesian-mechanics framing therefore belong in the curriculum as boundary-setting material, not as adversarial footnotes (Aguilera et al. 2022, 2023). Recent empirical reviews similarly motivate the platform's wording discipline: the curriculum can teach predictive coding and Active Inference as formal frameworks while still marking where direct empirical support is incomplete or domain-specific (Hodson et al. 2024). Psychology-and-psychiatry reviews add the same caution at the behavioral-modeling level: Active Inference is productive for model fitting and explanation, but its theory-level empirical adequacy still

requires direct comparison with alternatives (Badcock and Davey 2024). Gershman's critique is useful here because it asks what distinctive predictions the FEP contributes, which is exactly the kind of scope boundary this manuscript treats as a claim-support obligation (Gershman 2019). The perception literature supplies a complementary, narrower evidence map: Bayesian vision and predictive-processing accounts connect generative hypotheses to perceptual inference, while cortical-microcircuit, efficient-coding, perceptual-decision, and neurophysiological reviews expose model-specific assumptions and unresolved empirical questions (Yuille and Kersten 2006; Clark 2013; Hohwy 2013; Kersten et al. 2004; Bastos et al. 2012; Spratling 2017; Summerfield and Lange 2014; Wei and Stocker 2015; Millidge, Seth, et al. 2021; Smith et al. 2020; Seth and Hohwy 2020). These sources sharpen the perception-as-inference lane without turning predictive processing into

Evidence-status critiques and empirical boundaries I

a settled theory of the brain or this curriculum into a neuroscience intervention.

Information, variational, filtering, and application sources

Information and filtering sources I

The information-theoretic and variational strata follow standard treatments of entropy, KL divergence, variational inference, graphical-model message passing, the ELBO/free energy relationship, and rate-distortion or bottleneck views of task-relevant compression (Shannon 1948; Beal 2003; Cover and Thomas 2006; Tishby et al. 1999; Bishop 2006; Jordan et al. 1999; Winn and Bishop 2005; Blei et al. 2017). State-space filtering, continuous-time predictive coding, and neural message passing are represented in the curriculum, with deeper executable worked examples left for later releases; the relevant literature includes Bayesian filtering, Helmholtz machines, hierarchical predictive coding, the graphical-brain account, DEM, generalized filtering, and generalized free energy (Särkkä 2013; Dayan et al. 1995; Rao and Ballard 1999; Friston 2008; K. J. Friston, Parr, et al. 2017;

Friston et al. 2008; Friston et al. 2010; Parr and Friston 2019; Kataoka and Doya 2026; Tschantz et al. 2023). Probability, information, and Bayesian-analysis textbooks provide adjacent mathematical foundations for these implementations and help distinguish a reusable inference method from any one Active-Inference formulation (MacKay 2003; Jaynes 2003; Murphy 2012, 2022; Gelman et al. 2013).

Information and filtering sources II

Broader applications to embodied cognition, biological self-organization, collective behavior, action-oriented model learning, and institutional sensemaking motivate the applications and practice strata (Clark 2016; Friston 2013, 2019; Tschantz et al. 2020; Ramstead et al. 2018). Application-facing work spans cognitive control, hierarchical motivation, computational consciousness, clinical neuroscience, control-as-inference comparisons, robotics, and model-based cognitive flexibility (Pezzulo 2012; Pezzulo et al. 2018, 2023; Vilas et al. 2022; Smith et al. 2021; Imohiosen et al. 2020;

Millidge et al. 2020; Schwartenbeck et al. 2015; Lanillos et al. 2021; Catal et al. 2021; Queißer et al. 2021; Sales et al. 2019). These are application and comparison surfaces, not evidence that the present discrete kernels scale to robotics, clinical decision support, or human cognitive change. Predictive-learning neuroimaging evidence strengthens the plausibility of predictive learning as a representational mechanism, but it remains neural-task evidence rather than evidence that this microlearning curriculum changes learner representations (Greco et al. 2024). Current frontier sources sharpen the roadmap boundary. Reactive message passing and RxInfer show a software and algorithmic path for scalable Bayesian message passing; coupled free energy points toward robust variational geometry; active predictive coding and divide-and-conquer predictive coding extend the neural-inference and scalability context. Those sources are cited as scope-extending

scholarship, not as implemented kernels, production-scale evidence, or learner-outcome evidence in this release (Bagaev and Vries 2023; Bagaev et al. 2023; Nelson et al. 2025; Rao et al. 2024; Sennesh et al. 2024). Recent experimental work also grounds the FEP outside purely theoretical exposition (Isomura et al. 2023).